

LIEKSA LAMPELA, Finland

WMO: 027960

Lat: 63.321N

Lon: 30.046E

Elev: 98

StdP: 100.15

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.5	-26.4	-32.7	0.2	-29.5	-29.3	0.3	-26.4	7.5	-3.3	6.9	-3.3	0.3	100	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	26.7	18.0	24.6	17.1	22.7	16.5	20.0	24.3	18.7	22.7	17.5	21.3	2.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.5	22.1	17.1	12.3	20.7	15.9	11.4	19.1	57.8	24.2	53.3	22.7	49.6	21.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	6.1	5.4	-31.6	28.7	4.0	3.4	-34.4	31.1	-36.7	33.1	-39.0	35.0	-41.8	37.4		
			-31.3	21.0	3.9	1.9	-34.2	22.4	-36.4	23.5	-38.6	24.6	-41.5	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-10.9	-8.6	-4.9	1.5	8.8	13.0	16.2	14.3	9.5	3.0	-1.3	-6.0
	DBStd	10.39	8.13	7.56	5.63	4.03	4.54	3.85	3.43	3.35	3.40	3.63	4.75	7.11
	HDD10.0	3086	647	522	463	255	78	14	2	5	49	217	339	496
	HDD18.3	5644	906	755	721	504	298	164	83	131	264	474	589	754
	CDD10.0	513	0	0	0	1	39	105	193	138	36	1	0	0
	CDD18.3	28	0	0	0	0	1	4	17	6	0	0	0	0
	CDH23.3	328	0	0	0	0	31	63	176	59	0	0	0	0
	CDH26.7	56	0	0	0	0	2	8	36	10	0	0	0	0
Wind	WSAvg	2.4	2.0	2.3	2.6	2.3	2.6	2.6	2.2	2.2	2.3	2.7	2.6	2.4
Precipitation	PrecAvg	658	43	36	33	33	50	69	85	75	67	62	53	50
	PrecMax	873	76	62	64	67	99	132	160	165	134	138	118	91
	PrecMin	544	14	8	13	6	15	29	21	23	14	15	13	11
	PrecStd	82	16	16	12	13	22	26	39	34	30	27	22	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	4.0	8.0	16.7	26.4	27.8	29.8	27.9	20.1	12.4	8.7	4.8
		MCWB	1.9	2.5	3.5	9.8	14.7	17.7	21.2	19.6	15.7	11.2	7.8	4.4
	2%	DB	1.5	2.4	5.0	13.2	23.6	25.2	27.2	24.8	18.0	10.6	6.7	2.7
		MCWB	0.9	1.3	1.6	7.1	14.0	17.2	19.2	17.7	14.1	9.3	5.9	2.0
	5%	DB	0.5	1.3	3.1	10.5	20.7	22.7	25.1	22.4	16.3	9.5	5.5	1.8
		MCWB	0.2	0.7	1.1	5.4	13.1	15.6	18.2	17.0	13.5	8.3	5.0	1.2
	10%	DB	-0.2	0.6	2.0	8.2	17.7	20.5	23.2	20.5	15.0	8.4	4.0	1.0
		MCWB	-0.5	0.1	0.5	4.3	11.0	14.6	17.4	16.1	12.9	7.3	3.5	0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.6	4.1	10.2	16.4	20.1	22.2	21.0	16.6	11.6	7.9	4.4
		MCDB	2.6	3.6	7.0	15.3	23.5	24.3	28.1	26.6	19.3	12.3	8.5	4.7
	2%	WB	1.1	1.5	2.4	8.3	15.0	18.4	20.6	18.8	15.2	9.9	6.1	2.1
		MCDB	1.5	2.1	4.1	12.4	21.3	23.1	25.5	22.2	16.9	10.3	6.6	2.5
	5%	WB	0.3	0.8	1.5	5.9	13.5	16.7	19.4	17.7	14.0	8.5	5.1	1.3
		MCDB	0.6	1.2	2.8	10.0	19.5	21.2	23.7	21.4	15.8	9.2	5.5	1.6
	10%	WB	-0.6	0.2	0.8	4.6	11.8	15.3	18.3	16.7	13.0	7.3	3.6	0.7
		MCDB	-0.2	0.7	1.8	7.4	16.8	19.4	22.1	20.1	14.7	8.3	4.0	1.0
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	9.4	9.7	10.8	9.9	9.7	9.1	7.5	5.2	3.9	5.0
		MCDBR	3.5	4.6	8.4	14.0	16.3	13.6	13.3	12.8	10.0	6.6	4.6	3.4
		MCWBR	3.4	4.3	6.2	8.4	8.0	6.6	6.2	6.8	6.8	5.7	4.3	3.4
	5% WB	MCDBR	3.5	4.3	6.9	12.6	14.5	12.1	11.8	10.5	8.7	6.2	4.5	3.4
		MCWBR	3.5	4.1	5.4	7.8	7.6	6.3	6.1	6.0	6.2	5.6	4.3	3.4
Clear-Sky Solar Irradiance	taub	0.226	0.284	0.312	0.349	0.351	0.347	0.373	0.354	0.332	0.310	0.241	0.211	
	taud	1.984	2.100	2.157	2.280	2.444	2.484	2.426	2.465	2.513	2.390	2.075	1.966	
	Ebn at Noon	375	629	772	828	864	875	836	822	771	622	355	184	
	Edh at Noon	31	66	94	104	97	95	99	87	68	51	28	15	
All-Sky Solar Radiation	RadAvg	0.18	0.74	2.19	3.67	4.74	5.23	4.94	3.75	2.06	0.77	0.19	0.07	
	RadStd	0.02	0.17	0.26	0.39	0.54	0.34	0.51	0.41	0.23	0.12	0.04	0.01	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (25 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	-186	N/A	N/A	N/A	

Nomenclature: See separate page